Introduction

It is commonly known that there is a gender gap in terms of income - women earn about 82 to 88 cents for every dollar a man earns. While this statistic may be widely accepted, it often circulates without deeper examination of the systemic structures that produce it.

One such structure that may contribute to this gap is the gender distribution within academic fields. Different fields often correlate with different industries and, consequently, different salary levels. For instance, fields dominated by men, such as computer science or engineering, tend to lead to higher paying roles compared to female dominated fields, such as education or psychology. Some of this disparity is simply rooted in the field or department of study, regardless of personal skill or educational investment.

The paper aims to explore the relationship between the gender composition in academic departments and financial success post-graduation. This relationship can reveal whether there are systemic differences and inequalities in higher education that lead to long term economic disadvantages for students, particularly women. These findings intend to inform policymakers striving to address gender equality in STEM and other high earning sectors. Furthermore, this paper seeks to understand whether fields with a higher percentage of female students are less valued in the labor market, regardless of who enters them. This research moves beyond surface level statistics and delves into societal norms that may lead students towards very different financial futures based upon gender.

Existing Research

Existing research has explored elements of both gender and post-graduation outcomes, but limited work has been done to study the joint relationship between gender composition in

higher education departments and financial outcomes post-graduation. While related studies examine aspects of societal gender dynamics in education and gender wage disparities in the workforce, few studies directly connect these elements, particular in a robust manner.

Several studies aim to show how faculty gender representation may alter student success levels. For example, Mansour et al. (2020) examined the effect of the gender of STEM professors on female students' post graduate outcomes, concluding that gender diversity in professors leads to positive outcomes. While these outcomes were merely indicators of professional success rather than financial, they still demonstrate that gender may be a key factor in student success post graduation. Still, the study's external validation is limited, as it draws on data from a single, highly selective university (U.S. Air Force Academy), and therefore can not necessarily be generalized.

Extending upon a single institution, Boustan and Langan (2019) explored the differences in women's representation and outcomes across economic PhD programs. The findings echoed the importance of faculty gender diversity, determining that having a greater share of female faculty members contributed to high female success rates. However, females graduating from the same undergraduate economic programs as men tended to secure lower ranking PhD programs post graduation and be published less frequently in top journals. While this paper offers intriguing insights, it does not track wage data and instead focuses on other metrics of success. As such, it does not address whether gender diversity leads to differing financial success.

A broader structural lens is taken by Tookes and Sherman (2021), who found that between 2009 and 2017, women comprised only 16% of finance faculty at the top 100 U.S. business schools. These gender differences are stark - particularly in terms of salary, placement, and tenure even when compared to male counterparts at comparable universities, even when

accounting for numerous individual and institutional characteristics. Moreover, there are structural differences that contribute to these outcome gaps and more work must be done to better understand such differences.

In contrast to faculty-focused studies, further research has investigated the role of peer gender composition. Zölitz and Feld (2020) concluded that females randomly assigned to sections with a higher presence of female peers were more likely to choose male dominated majors such as Finance. Moreover, the influence of peer gender composition and gender diversity reduced gender segregation in education choices, which could affect earnings. However, the scope of the study is limited to major selection and does not investigate a further financial connection.

While the given studies are strong in that they identify gender or gender composition as a differentiator in student success, they largely overlook financial consequences. To better understand this component, we must turn to the literature on gender wage gaps. Vincenzo et al. (2019) utilized an OLS method to conclude that the wage gap varies significantly across industries and sectors, indicating that in some areas that disparities are significantly more pronounced. While the designs of the experiment were robust, including the control of numerous factors that may explain the wage gap, there is no connection to the educational environment.

Building upon this, Dong & Zhang (2024) analyzed evidence from university graduates in China, demonstrating that despite the fact that females tended to outperform males in their undergraduate career, female graduates faced significant wage gaps that widened overtime as the amount of time from graduation grew. In line with the findings of Vincenzo et al., the gender wage gap was more pronounced in certain majors, such as Finance and Management. However, its international context raises questions about if the findings can be generalized to the U.S.

These studies strongly suggest that gender plays a pivotal role in student outcomes. Furthermore, prior research on gender outcomes and financial success studies independently may be extensive, but there is yet to be relevant conclusions drawn about their joint relationship. This paper addresses that intersection. Furthermore, gender peer composition and diversity is studied sparsely, and must be more directly targeted to understand how systematic educational structures contribute to economic inequality.

Data Source

To complete this study, data was pulled from IPUMS USA (Integrated Public Use Microdata Series), which compiles over 60 high-precision samples created from sixteen federal censuses and the American Community Surveys. The specific data extract used responses from the years 2020 through 2023. This was to ensure recent, post-COVID years were represented. However, the inclusion of the year 2020 may present a limitation, as income and employment patterns varied drastically in that year, potentially distorting the results as compared to more stable years.

Furthermore, the initial dataset contained over 12.5 million entries. After filtering to include only individuals with a bachelor's degree or higher, approximately 3.5 million entries remained. Due to both the large sample size and the census-based data collection method, there is limited to no selection bias.

The dependent variable used in this analysis is INCWAGE, which reports the total pre-tax wage and salary income over the past year from the individual respondent. Notably, this variable contained no missing values in the filtered dataset.

Additionally, two existing variables were used to create the primary independent variable. These variables were SEX and DEGFIELD, which reports the field in which a respondent received their bachelor's degree. While sex is a discrete binary variable, the degree of the field variable offered more than 65 distinct categories. These two variables were merged to calculate the proportion of female respondents in each field of degree, which was then added back to the dataset. The IPUMS dataset is comprehensive in its available variables, facilitating the inclusion of several covariates to control for confounders factors

While missing values are typically ever-present in survey data, filtering the data by educational attainment and year excluded these missing values. Because both the independent and dependent variables had no missing values or codes corresponding to missing categories (ex: 9 in SEX means N/A), it is likely that IPUMS either imputed or excluded incomplete cases during extraction. This is important to acknowledge, as it may mean that the results are not fully representative.

Statistical Analysis and Assumptions

To understand if the gender makeup in academic departments or fields alters financial success level post graduation, a multivariate ordinary least squares regression model was used. This model was then analyzed to assess the main causal inference. Additionally, a log transformation was applied to the INCWAGE variable after analysis of the visual distribution and a Box-Cox test. This was done to ensure the results were more accurate and robust.

Numerous control variables were identified from an analysis of existing studies, in addition to personal reasoning. The variables were mapped in a DAG (Directed Acyclic Graph) to determine which controls were necessary (Appendix A). The DAG identified seven potential

confounding variables. Of these, five variables were controlled for in the final regression analysis.

The first confounding variable was socioeconomic status (SES). Individuals with lower SES may feel more constrained in their choice of major/field. Additionally, SES can predict income in numerous ways, such as through access to resources and professional networks. SES was controlled by adding the POVERTY variable to the regression. This variable represents a family's total income as a percentage of the federal poverty threshold, which serves as a good reflection of the respondent’s relative SES.

The next variable recognized as a confounder was gender. The gender makeup of a department is heavily dependent on personal gender preferences, such as wanting to be in a non-male-dominated degree. Additionally, there is strong evidence of a gender pay gap. This was accounted for by including the SEX variable in the regression in addition to the independent variable.

Next, geographic location was flagged as a potential source of bias. Geographic location can influence which majors are accessible or feasible, which could relate to the gender makeup of such a department. Additionally, being in a big city versus a rural environment can significantly influence earning potential. COUNTYICP, or the county where the household was enumerated, was added to the regression. The use of this variable poses a potential source of concern in practicality, as people may move frequently or go to university away from home.

Moreover, the DAG identified nationality as a potential confounder. Cultural norms about gender roles can vary across countries. Additionally, international students or immigrants may face disadvantages in the labor market, leading to reduced income. BPL was included to mitigate its potential bias, which identifies the U.S. state or foreign country where a respondent was born.

Additionally, race/ethnicity was identified as influencing both field choice and income. The intersectionality of race and gender can have significant implications in terms of representation within a given field. Additionally, there is strong evidence of racial disparities in income in the labor market. Adjustment for this factor was made by using two variables - RACE and HISPAN, as the inclusion of only the race variable would have been insufficient.

Two variables identified as confounders were not able to be controlled for in the regression analysis. The first of these variables is field prestige. Some fields offer characteristics that may be more appealing to women. For instance, being a teacher is sometimes appealing to women as it follows a short school day which allows time to spend with family and children, which some women value. Additionally, fields vary widely in average pay, regardless of gender. For example, finance versus education pays everyone differently regardless of gender. OCC, or occupation, was identified as the only suitable replacement for this, but was not used because it contained over 40% missing values.

Furthermore, the academic capabilities of an individual was identified as a control variable, but was not controlled because after extensive search through variables and codebooks, it was determined that there were no available measures in the dataset that captures student ability or motivation.

Within the multivariate regression model, several assumptions were tested. First, a DFBETA test was run to test for influential cases (Appendix B). Since the range on the plots remained between -1 and 1, it can be concluded that no one single observation has an extreme influence on the model's estimates. Furthermore, a Variance Inflation Factor (VIF) test was used to examine multicollinearity (Appendix B). Since all the VIF values were below a threshold of 5, it can be reasonably concluded that the predictors are independent of each other. Lastly, a

Jarque-Bera test was used to test for normality of residuals (Appendix B). Because the p-value from this test was very small ($p < 0.05$), the results are statistically significant, indicating that the residuals are not normally distributed. Additionally, the Q-Q plot revealed a notable S-shaped pattern, which deviates from the theoretical linear line. Therefore, the normality assumption is violated, which must be considered when evaluating the robustness and application of the results.

Results

OLS Regression of Log Income on Gender Composition and Controls

	Bivariate Model	Multivariate Model
Intercept	9.042***	4.297***
	(0.008)	(0.014)
Percent Female in Field	-0.028***	-0.019***
	(0.000)	(0.000)
Poverty		0.011***
		(0.000)
Sex		-0.330***
		(0.006)
County		0.000***
		(0.000)
Place of Birth		-0.000
		(0.000)
Race		0.144***
		(0.001)
Hispanic Origin		0.164***
		(0.004)
Num.Obs.	3412126	3359128
R2	0.025	0.089
R2 Adj.	0.025	0.089

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors in parentheses.

In addition to the table, the F-statistic p-values were calculated for both the bivariate and multivariate models, both of which were less than $2.2e^{-16}$. Therefore, both of the models are statistically significant in helping to explain income. From a closer inspection of the individual p-values within the multivariate regression model, every control variable except place of birth (BPL) was significantly significant (very small p-values). This means that all of the confounding variables help to explain the dependent variable, apart from birthplace.

The coefficient of the main independent variable in the multivariate regression is -0.019. This means that as the percentage of female individuals in a field increases by 1 percentage point, the total log income from wages within the last year decreases by 0.019%, or 0.981 dollars ($e^{-0.019}$). This represents a decrease in income by approximately 1.88% ([1 - 0.981] x 100), or a change to 98.1% of the previous income level. Since the coefficient is negative, this demonstrates an inverse relationship: as the percent of women in a field increases, income decreases slightly.

To contextualize the practical impact, the interquartile range was examined. Multiplying the multivariate independent variable effect size by the IQR results in a 0.547 decrease in log income, which corresponds to an estimated 42.15% difference in income. Furthermore, compared to the total range of income, this shift accounts for approximately 4% of the total range (0 to 13.68), representing a substantial structural difference attributed to field level gender composition

The comparison of the bivariate and multivariate models is not surprising, as some of the initial effect in the bivariate model was due to confounders, which were accounted for in the second model. The coefficient still remains negative and meaningful, indicating that even after

accounting for confounding variables, a higher percentage of women in a given field is associated with lower income.

Additionally, the R-squared value increased from 2.5% to 8.9%, indicating that a greater portion of the variance in income is explained by the model once confounding variables are added. This number is still relatively small though, indicating that a large portion remains unexplained. This is somewhat expected when analyzing income, as there are several unobserved and unmeasurable factors, such as motivation and educational opportunity, that likely contribute to differences in earnings.

In order to make the model more robust, two nonlinear terms of order two and three, in addition to one interaction term, were added to the model. The first nonlinear term was poverty, as it relates to money and finances, which are commonly nonlinear or skewed in nature. The second nonlinear set of terms involved percent female, as it is critical to examine the main independent variable more closely. Furthermore, an interaction between poverty and percent female was tested to examine whether the effects of poverty on income varies by gender composition. This analysis was motivated by the possibility that fields with more women may have different barriers to entry or may be more affected by occupational segregation and gender discrimination, potentially altering the relationship between poverty and income.

The intercepts were plotted for the nonlinear terms to ensure that the addition of the variable would be productive (Appendix C). Additionally, an appropriate test was conducted to evaluate the sufficiency of the interaction term (Appendix C). After all new terms were added to the model, the results can be found in Appendix D. Within this new model, the p-value was still less than $2.2e^{-16}$, indicating the results are statistically significant. Additionally, the R-squared increased to 9.3%, demonstrating that an even greater portion of the variance is explained by the

model. The individual effect sizes were not analyzed in this study, as they were beyond scope and would require further analysis.

Conclusion, Limitations, and Next Steps

From an analysis of a multivariate regression model used to predict income from wages within the last year from the proportion of female students in a given field in addition to controls, a causal relationship is determined. Furthermore, as the percentage of female students increases, the total income decreases, specifically by 1.88% compared to the previous income level.

This points toward the notion that fields that are dominated by women inherently generate lower wages. There are numerous theories as to why this is. One such theory is that females are not encouraged to enter male-dominated fields that pay more. This has strong policy implications, suggesting that more work must be done in order to encourage females to pursue any field they wish. A competing theory is that the labor market systematically devalues work associated with women. As more women choose to enter a field, society may come to value the field less, perceive its quality as lower, or assume the work is easier - a concept referred to as "occupational feminization". This is due to socialized norms regarding attributes associated with women. This may be more difficult to tackle, as it requires not only supporting women in accessing higher-paying fields, but also revealing how we reward and value different types of work in society as a whole.

However, still much of the variance in the dependent variable remains unexplained by the regression model, suggesting that more variables are needed or that something key is missing. This may be due to the two confounding variables that were identified earlier but excluded, as no suitable proxies were available in the dataset. This limits the conclusions of the study, and future

research should aim to include field prestige and academic capabilities as predictors to increase robustness.

Furthermore, these findings could be studied in conjunction with the widely cited statistic that women make 80 cents for every dollar a man makes. By bridging this common narrative and the present analysis, meaningful conclusions could be drawn.

Additionally, future work should look beyond binary gender dynamics. The present dataset only included options for men and women, but it would be valuable to contextualize these findings beyond this binary by examining how people with different identities (such as transgender, non-binary, and gender nonconforming) may encounter additional barriers.

Finally, the independent variable, department gender composition, serves as a proxy to represent field level gender dynamics, which could potentially be flawed. This variable does not capture the full complexity of the labor market patterns and societal expectations, such as differences in career support or workplace discrimination. These factors may play a significant role in professional and financial outcomes that may not be reflected in the present analysis. Future research should aim to develop a more comprehensive approach to representing gender composition and gender dynamics in quantitative models.

Despite these limitations, this study successfully establishes a clear, inverse causal relationship and lays important groundwork for continued research.

Work Cited

Alfano, V., Cicatiello, L., Gaeta, G. L., & Pinto, M. (2019). *The gender wage gap among PhD holders: An empirical examination based on Italian data* (GLO Discussion Paper No. 393). Global Labor Organization (GLO). https://hdl.handle.net/10419/202641

Boustan, L. P., & Langan, A. (2019). *Variation in women's success across PhD programs in economics* (NBER Working Paper No. 25444). National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w25444/w25444.pdf

Mansour, H., Rees, D. I., Rintala, B. M., & Wozny, N. N. (2020). *The effects of professor gender on the post-graduation outcomes of female students* (NBER Working Paper No. 26822). National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w26822/w26822.pdf

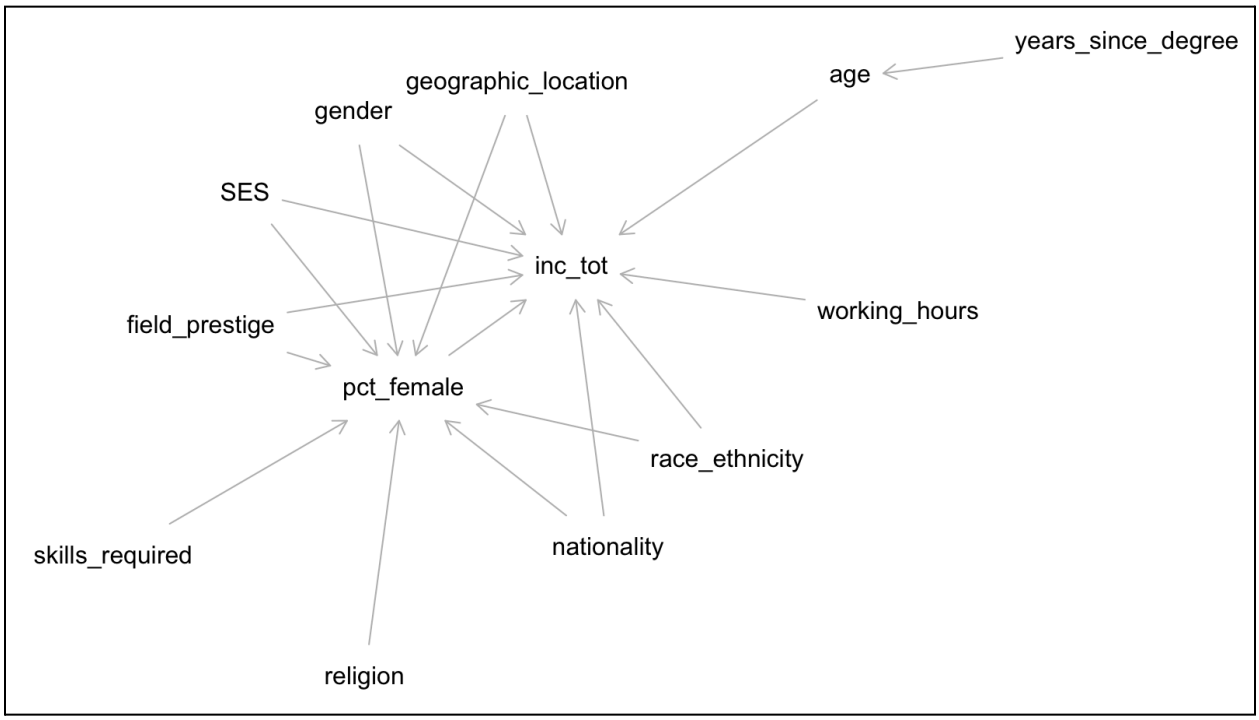
Ruggles, S., Flood, S., Goeken, R., Grover, J., Meyer, E., Pacas, J., & Sobek, M. (2023). *IPUMS USA: Version 13.0 [Dataset]*. IPUMS. https://doi.org/10.18128/D010.V13.0

Tookes, H. E., & Getmansky Sherman, M. (2021). *Female representation in the academic finance profession*. SSRN. https://ssrn.com/abstract=3438653

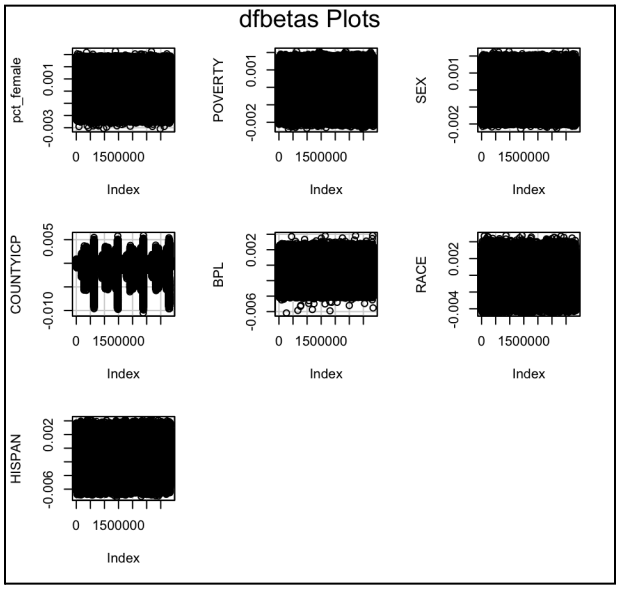
Zhang, Y., & Dong, Y. (2024). *Gender wage gap in early careers: Evidence from university graduates in China*. *Review of Development Economics*. Advance online publication. https://doi.org/10.1111/rode.13181

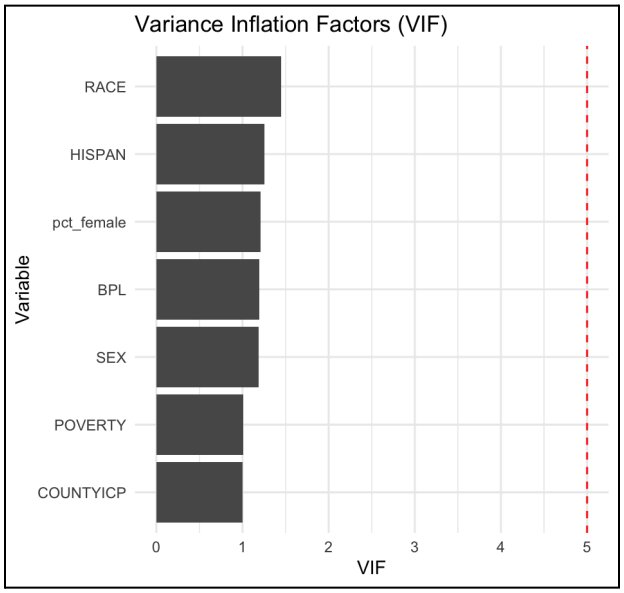
Zölitz, U., & Feld, J. (2020). *The effect of peer gender on major choice in business school* (IZA Discussion Paper No. 13396). IZA Institute of Labor Economics. https://docs.iza.org/dp13396.pdf

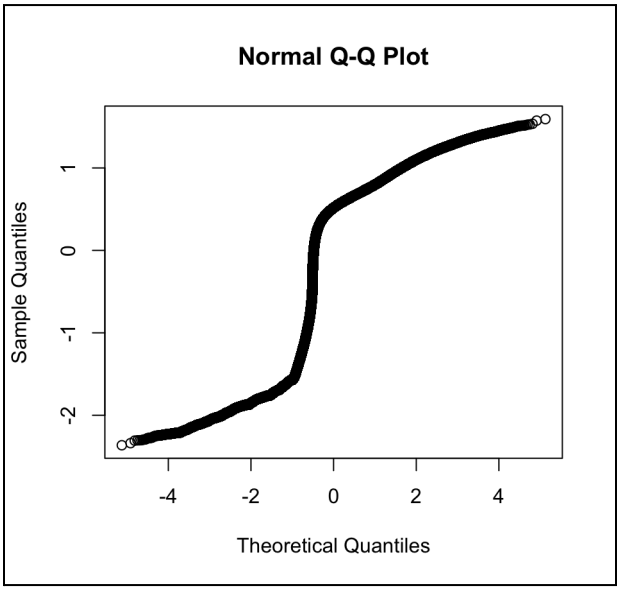
Appendix A


years_since_degree
age
geographic_location
gender
SES
inc_tot
working_hours
field_prestige
pct_female
race_ethnicity
skills_required
nationality
religion

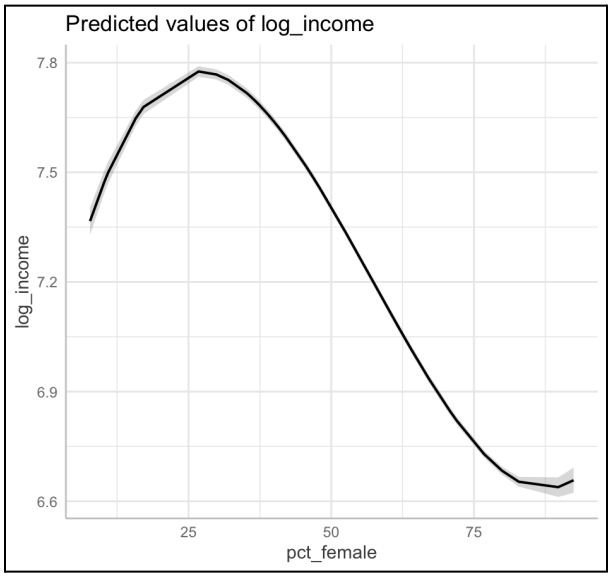
Appendix B


dfbetas Plots
pct_female
POVERTY
SEX
COUNTYICP
BPL
RACE
HISPAN
Index


Variance Inflation Factors (VIF)
RACE
HISPAN
pct_female
BPL
SEX
POVERTY
COUNTYICP
Variable
VIF


Normal Q-Q Plot
Sample Quantiles
Theoretical Quantiles

Appendix C


Predicted values of log_income
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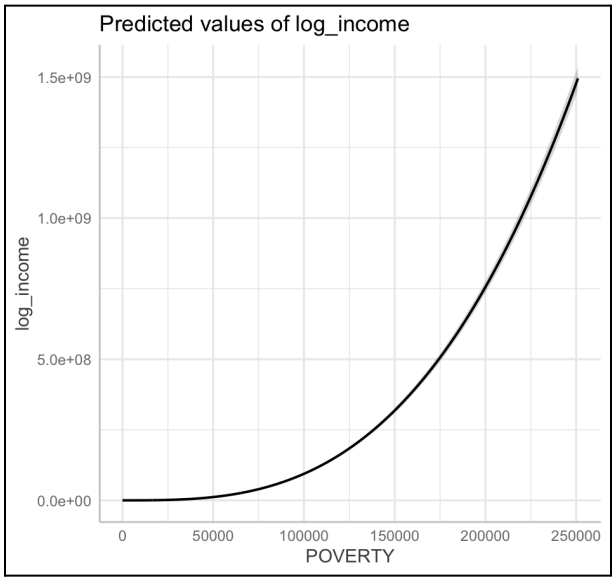
7.5
7.2
6.9
6.6
25
50
75
pct_female


Predicted values of log_income
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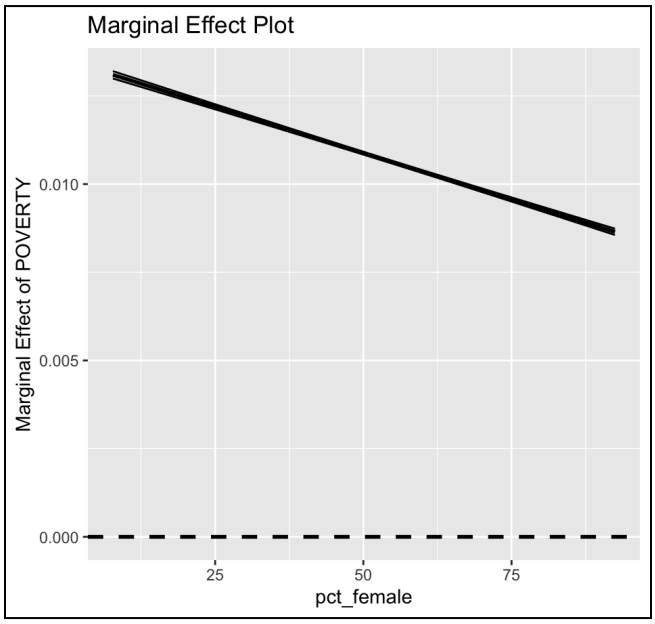
5.0e+08
0.0e+00
0
50000
100000
150000
200000
250000
POVERTY


Marginal Effect Plot
Marginal Effect of POVERTY
0.010
0.005
0.000
25
50
75
pct_female

Appendix D

OLS regression of Log Income on Gender Composition, Controls, and Advanced Terms

	Advanced Multivariate Regression Model
Intercept	-0.051
	(0.046)
Percent Female in Field	0.099***
	(0.002)
Poverty	0.039***
	(0.000)
Sex	-0.329***
	(0.006)
County	0.000***
	(0.000)
Place of Birth	0.000**
	(0.000)
Race	0.144***
	(0.001)
Hispanic Origin	0.160***
	(0.004)
I(pct_female^2)	-0.002***
	(0.000)
I(pct_female^3)	0.000***
	(0.000)
I(POVERTY^2)	-0.000***
	(0.000)
I(POVERTY^3)	0.000***
	(0.000)
Female % × Poverty	-0.000***
	(0.000)
Num.Obs.	3359128
R2	0.093
R2 Adj.	0.093

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Standard errors in parentheses.